Received: May 06, 2023 | Accepted: May 29, 2023 | Published: June 05, 2023

Crossref | PEERCITE PUBLISHERS Open Access

Peercite Journal of Cardiobiology

ORIGINAL RESEARCH

Review of Tricuspid Regurgitation in His Bundle Pacing: A Comprehensive Analysis

Jahanzeb Malik

Department of Cardiology and Clinical Physiology, Karolinska University Hospital, Sweden

Citation: Malik J (2023) Review of Tricuspid Regurgitation in His Bundle Pacing: A Comprehensive Analysis. Peercite J Cardiobiol. 1(2):1506-1511.

*Corresponding Author : Jordanna Santos Monteiro

Email :jordanna.santosmonteiro@gmail.com

Copyright: © 2023 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://peercite.org).

Abstract:

Objective: This systematic review aimed to investigate the association between novel TR and its quantification in patients undergoing His bundle stimulation (HBP).

Methods: A literature search was conducted using mesh terms (e.g. His bundle stimulation, tricuspid regurgitation, tricuspid regurgitation) in PubMed, EMBASE, Web of Science CINAHL, and the Cochrane Library up to October 2021. did. Relevant studies evaluating tricuspid regurgitation in HBP were included, and information on TR and its associated factors (ejection fraction (EF) and New York Heart Association (NYHA class)) were retrieved from eligible studies. 4,444 Results: Of 196 articles, 10 studies with 546 HBP patients met inclusion criteria. The mean age of the patients ranged from 61.2 ± 12.3 years to 75.1 ± 7.9 years, and 54.1% were male. The overall implant success rate was 79.2%. Only one study reported a 5% TR incidence, whereas nine studies reported no new TRs after HBP. Four studies reported an overall decrease in TR of 1 degree, and three studies showed an increase in TR from baseline. Two studies showed no change from baseline TR.

Conclusion: HBP leads to improved TR grade after HBP in cardiac resynchronization therapy (CRT) and atrioventricular block (AVB). Further studies in the form of randomized controlled trials are needed to further evaluate the impact of HBP on tricuspid valve function.

Keywords : Atrioventricular block, Ejection fraction

Introduction :

Approximately 1 million cardiac implantable electronic devices (CIEDs) have been implanted in patients worldwide for a variety of indications. The most commonly implanted CIED is a pacemaker. Since the introduction of the world's first cardiac pacemaker (1958), these devices have become the main treatment for bradyarrhythmias, and their indications continue to increase. Tricuspid regurgitation (TR) caused by cardiac implanted electronics is a cause of valve insufficiency or is exacerbated by right ventricular (RV) leads after pacemaker implantation. Right ventricular pacing leads are associated with poor prognosis. Regarding causation, it is difficult to distinguish between the role of mechanical disruption of the tricuspid valve by pacemaker leads and dyssynchrony caused by pacing, although it is possible that both contribute.

Major risk factors for CIED-related TR include implantable cardiac cardioverter defibrillator (ICD) leads, multiple ventricular leads, lead position relative to the tricuspid valve (TV), and inappropriate pacing position. Several studies have shown that ICD and CRT electrodes are associated with higher TR than pacemakers. The reason is that the electrodes are thick and have low elasticity. Over the past two decades, large randomized trials have investigated the role of biventricular pacing CRT (BiV-CRT) [1].

The BLOCK-HF trial showed significant reductions in overall mortality, heart failure requiring hospitalization, and >15% increase in left ventricular (LV) systolic volume index, favoring BiV-CRT over RV pacing.

In the BioPace trial, there was no significant trend toward BiV-CRT for the composite endpoint of mortality and first heart failure hospitalization (HR 0.87, p = 0.08). However, BiV-CRT is associated with increased lead-induced TR and is not an ideal strategy for patients with concurrent tricuspid regurgitation, among other challenges and contraindications. His-bundle pacing (HBP) provides efficient physiological and synchronous ventricular activation through direct stimulation of the His-Purkinje system, thereby mitigating the adverse effects of asynchronous ventricular activation that occurs with RV pacing. HBP can also normalize conduction in patients with bundle branch block (BBB), thus serving as an alternative to CRT. Since the HBP electrode can be implanted above the tricuspid annulus, it has minimal impact on TV morphology and results in TR reversal. In this systematic review, we present the association between TRs in patients undergoing HBP [2].

Materials and methods:

Search strategy:

This systematic review is based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement. Electronic databases including PubMed, EMBASE, Web of Science, CINAHL, and Cochrane Library databases were searched to identify key references. This review includes all eligible English studies. The following search terms were used: ("nerve bundle pacing [mesh]" OR "nerve bundle pacing [mesh]" OR "artificial cardiac pacing [mesh]" OR "parahistopathic pacing [mesh]" OR "physiological Pacing" OR "Conduction System Pacing" OR "Tricuspid Insufficiency [Mesh]" OR "Tricuspid Insufficiency [Mesh]" OR "Tricuspid Insufficiency [Mesh]" OR "Tricuspid Insufficiency [mesh]"). The authors also searched all review reference lists and Google Scholar in relevant fields to identify additional relevant studies. A total of 196 studies were initially identified by his two reviewers (D.I.S. and S.M.J.Z.) [3].

Study Selection and Results to avoid duplicate studies, a web-based software platform (Covidence) was used. Two reviewers (D.I.S. and S.M.J.Z.) then independently reviewed the title and abstract of the paper. In case, there was a disagreement between the two reviewers, and a third investigator (J.M.) contributed to resolving the impasse. Search the full text for possible relevance to determine if an article is included. The inclusion criteria were: randomized controlled trials (RCTs), observational studies, and case series (≥15 patients) reporting the incidence of TR after HBP before and after follow-up. Study endpoints include: (1) Quantification of TR degree at baseline and post-HBP follow-up.

Data Extraction:

Two reviewers (D.I.S. and S.M.J.Z.) independently extracted data from the included studies. (1) Basic information about the study, author name, year of publication, and study design. (2) baseline characteristics such as age, sample size, gender, HBP success rate, and follow-up; (3) Outcome data including parameters used in assessing TR severity, incidence of TR worsening or improvement, EF, and NYHA class after HBP. (4) Risk factors associated with worsening TR (if any). Data extraction was performed based on mutual consent, and any discrepancies were resolved by her third researcher (J.M.). If a study recorded multiple follow-up periods, the longest period was selected for analysis [5].

Outcome Measures:

The primary outcome was assessment of her HBP or TR grade of new TR after transplantation. The secondary outcome was assessment of QRSd.

Quality Assessment:

The quality of each study was assessed using the National Institutes of Health (NIH) Quality Assessment Tool for RCTs and Observational Studies (NIH, 2021). Studies were rated as poor on a score of 0 to 4, fair on a score of 5 to 7, and high quality on >7.

Definition of Terms:

According to the American Society of Echocardiography, TR is classified as none or minimal, mild, moderate, or severe. The severity of TR was classified as grade I (none or trace), grade 2 (mild), grade 3 (moderate), and grade 4 (severe). A 1-grade change in TR was defined as a worsening or improvement of ≥1 grade in TR grade after HBP compared to baseline during the follow-up period [6]. A transition from moderate to more than severe to a study-specific grade was considered a 1-grade change to ensure standardization across studies and to be consistent with American Society of Echocardiography grading. The same goes for improving the TR grade.

Statistical Analysis:

Statistical Package for the Social Sciences (SPSS) version 26 (IBM Corp, Armonk, NY, USA) was used for statistical analysis. Quantitative variables were expressed as mean and standard deviation (SD). Qualitative variables were expressed as frequencies and percentages [7].

Results:

Baseline characteristics:

After excluding duplicates, a total of 196 articles were retrieved using the search strategy, and 10 articles met the criteria [8]. Baseline study characteristics and levels of TR, EF, and NYHA classes at baseline and follow-up. The included studies were published between 2017 and 2021. This systematic review included 2 RCTs and 8 observational studies of 546 patients who underwent HBP implantation either alone or after RV lead placement (figure 1) [9].

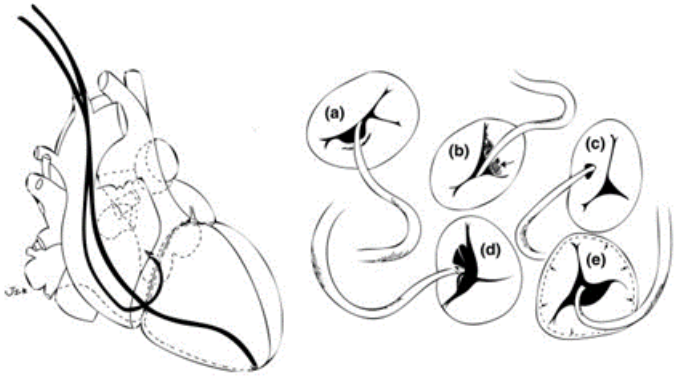

Figure 1: Mechanism of mechanical tricuspid regurgitation in the setting of permanent pacemaker.

That bundle pacing was used in three studies of CRT, five studies of AVB, one study of device upgrade to HBP, and one study of HBP generator modification. Baseline patient characteristics were described in his 10 papers from 546 patients [10,11].

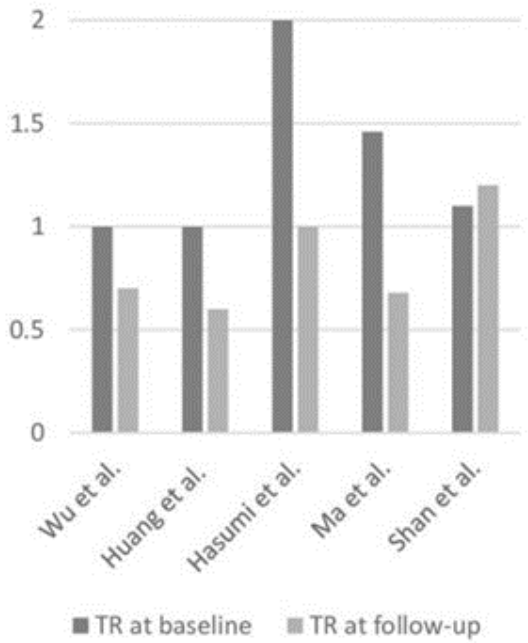

Figure 2: Tricuspid regurgitation grade at baseline.

The number and type of baseline characteristics reported varied between studies. The mean age ranged from 61.2 ± 12.3 years to 75.1 ± 7.9 years, and 54.1% were male. The transplant success rate was 79.2% (range, 51%-90%). Atrial fibrillation rates in HBP patients ranged from 11 to 60%. Six studies reported HBP electrodes implanted above the tricuspid annulus, and two studies reported implantation below the annulus (figure 2) [12].

Quality Assessment:

Primary endpoint:

Reporting rates for the target endpoint in the 10 included studies ranged from 0% to 5% (incidence of new TRs), with 9 studies reporting no new TRs. The incidence was not reported. Regarding the assessment of TR severity, three studies reported a decrease in overall TR severity after HBP implantation for cardiac resynchronization reported a mean TR grade of at least 1 at the start of the study (figure 3) [13].

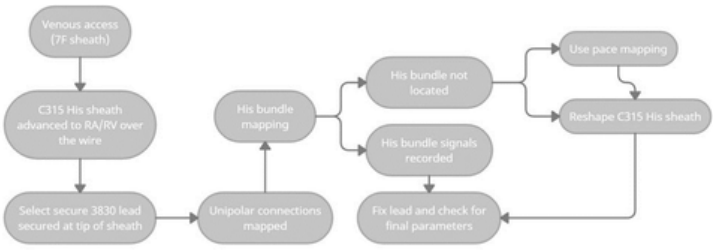

Figure 3: Algorithm for His bundle pacing**.**

Discussion:

We used an evaluation of TR consisting of 9 original research articles and 1 case series reporting approximately 5 years of experience with 546 patients at institutions worldwide. We systematically reviewed studies on HBP. This is the first systematic analysis of a patient cohort showing the incidence of TR with HBP and quantification before and after post-procedure follow-up and demonstrating its impact on ventricular synchrony.

Our systematic review shows that the risk of developing new TR after HBP is negligible (0%-5%) and improves TR grade. Although the prevalence of TR varied widely in these studies because the TR scoring system was not consistent across studies, the collected experience reports suggest that HBP is a viable option for patients with symptomatic TR. potential, indicating that HBP should be considered in a high proportion of patients requiring treatment. Ventricular pacing. Therefore, in the present study, we discussed the prevalence of TR after HBP transplantation to improve physicians' understanding of TR.

The studies included in this systematic review provide insight into who may be the target patient for HBP. The majority of patients who underwent HBP had baseline characteristics consistent with the typical pacemaker population. In contrast, two studies included patients with symptoms of heart failure and established indications for CRT, but the use of CRT as an alternative to apical RV pacing or BiV pacing was It is of great interest to the patient population. Recent evidence suggests a role for HBP in bundle branch block and left ventricular dysfunction, and as it can cause QRS stenosis in distal His bundle pacing, it is a candidate for CRT. However, long-term comparisons should be interpreted when clinical data from prospective clinical trials are available.

Mechanism of lead-mediated TR:

The phenomenon of lead-mediated TR was discovered in 1969. When this patient presented his TR murmur after his RV electrode implantation. When I disconnected the line, the noise disappeared. Based on this mechanistic understanding, many studies have discussed the incidence, mechanism, diagnosis, and treatment of lead-mediated TR. After electrode placement, TR can occur via multiple mechanisms. This can be caused by mechanical causes such as scarring or thrombus formation on the electrode, which can affect closure. Cracks and holes are another cause of TR. However, the most common mechanism in conduction-mediated TR is asynchrony, leading to aberrant RV activation by the cardiac pacemaker. Postmortem studies in the 1970s showed that pacemaker leads could attach to her TV leaflets or papillary muscles. In chronic TR, many studies have shown that adhesion of the TV cusp to the pacemaker lead causes movement restriction, resulting in abnormal coaptation of the posterior leaflet with the anterior and septal leaflets. In early experience with HBP, electrode placement into the HB was performed using a standard catheter. With the advent of modern tools and delivery systems, it is now possible to map HBs using the electrodes themselves, significantly increasing implantation success rates. A speculative advantage in HBP may be correct AV septal stimulation, which may prevent RV conduction-mediated TR. One study investigated the feasibility and safety of HBP in patients with atrial fibrillation. Apart from the primary objective, they showed no change in TR and mitral regurgitation (MR) at 6-month follow-up after HBP implantation. Several studies have shown that atrial fibrillation is a risk factor for developing tricuspid regurgitation. However, the above study showed a low incidence of TR after HBP implantation in patients with AF. Another study compared her HBP stimulation and her BiV stimulation in CRT candidates. This study showed that after HBP, baseline TR decreased from baseline at 12-month follow-up, as decreased LVEF due to heart chamber dilatation is a risk factor for the development of TR. Similarly, an abstract published in Circulation showed significant improvement in TR after HBP in all patients ($p = 0.033$) and in the subgroup with decreased LVEF ($p = 0.019$). Another study has shown the long-term consequences of HBP in patients with heart failure and LBBB. In addition to improvements in LVEF and QRSd, TR also improved from grade 3 to grade 1.

These studies demonstrated that low LVEF is a risk factor for the development of TR and that LV asynchrony caused by RV pacing may lead to a decrease in her LVED. Her RV pacing at the apex is a known cause of ventricular dyssynchrony that can lead to PICM. According to the literature, there is a 20% risk of developing PICM when the RV pacing load is 20% or more. By exploiting the His-Purkinje system, HBP can restore ventricular synchrony and reverse PICM. This systematic review shows not only that the likelihood of new conduction-mediated TR after HBP is negligible, but also that TR extent and EF improve significantly from baseline and follow-up. Although the included studies did not investigate the cause of the improvement in TR degree, based on the above principles, there appears to be a reversal of TV pathology caused by RV stimulation. Therefore, HBP may be an option for patients who develop lead-mediated TR.

Standard Techniques and Challenges:

A simplified algorithm for HBP injection. Currently, the most commonly studied electrode is the Select Secure 3830 (Medtronic). Implantation is done through an introducer sheath or catheter, such as the C315 HIS or the newly approved Select Site C304 HIS catheter. The C314 HIS introducer sheath has a 7.0F outer diameter and distal (primary) and proximal (secondary) curvatures to guide toward the tricuspid annulus. After venous access, introducer sheath is advanced over the guidewire to the TV annulus.

The 3830 electrode is then advanced to the tip to image the HB region in a monopolar configuration. The system is maneuvered towards the AV septum. Once the HB ECG has been identified and the operator is satisfied with the stimulation parameters of his HB range, he secures the electrode by rotating the electrode body clockwise. Flipping or rotating the electrode is usually an indication that it is properly secured. The sheath is then withdrawn into the right atrium (RA) where electrode parameters are evaluated in both monopolar and bipolar configurations. Implanting the electrode distal to the disease is important to achieve a low capture threshold.

Future Outlook and Limitations:

Despite promising results with HB mapping, 20% of patients have more distal or extensive conduction defects, and HBP can reach BBB with appropriate implantation and capture thresholds. may not be coveredF. introduced a technique to stimulate the left inferior cruciate ligament via an intraseptal approach to restore function of the distal Purkinje conduction system. However, more data are needed to evaluate the safety and efficacy of this technology. Therefore, LBBB pacing is a complementary option for patients with distal conduction disease and ventricular dyssynchrony. Regarding future research, despite the above data, there are no large randomized trials comparing the effectiveness of HBP with the effectiveness of conventional HER RV pacing. Evaluate and quantify TR and its mechanisms in HBP. The two included RCTs show a negative incidence of TR, but there are no data on long-term outcomes. Additionally, large multicenter randomized trials comparing HBP and BiV stimulation to assess EF and NYHA class are needed to evaluate the clinical efficacy of HBP. A registry should be established for patients with HBP, including but not limited to TR and other echocardiographic parameters regarding LVEF and RV function.

Finally, improving HBP success rates requires improved lead design and delivery tools in addition to data. This systematic review has several limitations. First, most of the included studies did not have outcome measures to quantitatively identify TR. Therefore, a quantitative synthesis could not be performed because the TR scoring system was not standardized across different studies. Therefore, relevant data regarding quantitative endpoints of TR need to be further improved. Second, most of the included studies were not randomized and confounders were not controlled. The observational nature of most studies introduces inherent limitations that may influence the observed results. Third, these studies lacked data that could lead to worsening or improvement of TR, such as: Examples: pulmonary hypertension, mitral regurgitation, enlarged RV size, etc. Fourth, there was heterogeneity among studies regarding the indications for HBP. Finally, more controlled and randomized trials should be conducted to evaluate the reach and long-term effects of TR with and without HBP.

Conclusion:

In summary, HBP represents just the most physiological form of ventricular activity, and current data show that HBP improves the degree of TR in patients implanted with CIED or candidates for CRT. suggests that it leads to However, this technique needs to be widely applied, and large randomized trials could help further validate its effectiveness.

Conflict of interest:

Author declares there is no conflict of interest.

References:

1. Abdelrahman M, Subzposh FA, Beer D et al. (2018) Clinical outcomes of his bundle pacing compared to right ventricular pacing. J of the A College of Cardio. 71(20):2319-2330.

2. Addetia K, Harb SC, Hahn RT et al. (2019) Cardiac implantable electronic device Lead-induced tricuspid regurgitation. JACC: Cardio Imaging. 12(4):622-636.

3. Arabi P, Özer N, Ateş AH et al. (2015) Effects of pacemaker and implantable cardioverter defibrillator electrodes on tricuspid regurgitation and right sided heart functions. Cardio J. 22(6):637-644.

4. Barba-Pichardo R, Manovel Sánchez A, Fernández-Gómez JM et al. (2013) Ventricular resynchronization therapy by direct his-bundle pacing using an internal cardioverter defibrillator. Europace. 15(1):83-88.

5. Curtis AB, Worley SJ, Chung ES et al. (2016). Improvement in clinical outcomes with biventricular versus right ventricular pacing: The BLOCK HF study. J of the A College of Cardio. 67(18):2148-2157.

6. Dandamudi G, Vijayaraman P (2016) How to perform permanent his bundle pacing in routine clinical practice. Heart Rhythm. 13(6):1362-1366.

7. Furman, S., & Schwedel, J. B. (1959 Nov). An intracardiac pacemaker for stokes-Adams seizures. The New England J of Med. 5(261):943-948.

8. Gibson TC, Davidson RC, DeSilvey DL (1980) Presumptive tricuspid valve malfunction induced by a pacemaker lead: A case report and review of the literature. Pacing and Clin Electrophysiology. 3(1):88-95.

9. Hu Y, Li H, Gu M et al. (2021) Comparison between his-bundle pacing and left bundle branch pacing in patients with atrioventricular block. J of Interventional Cardi Electrophysiology. 62(1):63-73.

10. Huang W, Su L, Wu S et al. (2019). Long-term outcomes of his bundle pacing in patients with heart failure with left bundle branch block. Heart. 105(2):137-143.

11. Nachnani GH, Gooch AS, Hsu I (1969) Systolic murmurs induced by pacemaker catheters. Archives of Internal Med. 124:202-205.

12. Qian Z, Zou F, Wang Y et al. (2019). Permanent his bundle pacing in heart failure patients: A systematic review and meta-analysis. Pacing and Clin Electrophysiology. 42(2):139-145.

13. Wey HE, Desai PV, Lattell J et al. (2019) Abstract 15104: His-bundle pacing improves tricuspid regurgitation in patients with bradyarrhythmia. Circulation. 140:A15104.